

Research Report: Detailed Specifications of Electric Seat Motors in Modern Battery Electric Vehicles (2024-2026)

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Objective: This report provides a comprehensive analysis of the technical specifications, operational characteristics, and market landscape for electric seat motors used in Battery Electric Vehicles (BEVs), with a specific focus on European and Asian models sold in the European market for the 2024-2026 period.

Executive Summary

The proliferation of Battery Electric Vehicles (BEVs) has catalyzed significant advancements in automotive component technology, including the small but critical DC motors that power adjustable seating systems. As vehicle interiors evolve into more sophisticated, comfortable, and personalized spaces, particularly within the quiet cabin environment of a BEV, the performance of these motors has come under greater scrutiny. This report details the specifications of these motors, revealing a market driven by demands for higher efficiency, lower noise, reduced weight, and increased functionality. Analysis of leading suppliers like Bosch, Brose, Nidec, and Johnson Electric indicates a clear technological trajectory [5, 17, 24]. Standard BEV models typically employ a limited number of cost-effective brushed DC motors for basic adjustments [5, 19]. In contrast, premium models feature a greater quantity of advanced, quieter Brushless DC (BLDC) motors to support complex, multi-way adjustments, massage functions, and integrated memory systems [5, 16, 18]. Key specifications for these 12V DC motors include current draws ranging from under 3 amperes at no-load to over 12 amperes at stall [32, 41], power consumption typically between 40 and 80 watts per motor [15, 32], and high-torque outputs achieved through sophisticated gearing [32]. The overarching trend is a move towards integrated, intelligent mechatronic systems that enhance passenger comfort and contribute to the overall efficiency and luxury experience of modern BEVs.

1.0 Introduction to Automotive Seat Motor Technology in BEVs

The electric seat motor, typically a small direct-current (DC) motor, is a fundamental component of modern automotive interior design, responsible for the automated adjustment of a vehicle's seats [62]. In the context of Battery Electric Vehicles (BEVs), the role and characteristics of these motors are evolving in response to new technological paradigms and heightened consumer expectations. The primary function of these motors is to convert electrical energy from the vehicle's low-voltage system into mechanical force, enabling a range of movements such as forward and backward sliding (fore/aft), height adjustment, backrest recline, and seat cushion tilting [11, 19]. In more advanced applications, they also power features like lumbar support, adjustable side bolsters, headrest positioning, and even massage functions [11, 19].

The importance of these motors extends beyond mere convenience. Proper seat positioning is critical for driver ergonomics, visibility, and safety, ensuring optimal posture and control. In the era of BEVs and the concurrent development of autonomous driving, vehicle interiors are being reimaged as flexible, multi-purpose environments for work or relaxation. This shift necessitates more dynamic and ad-

adjustable seating configurations, which in turn relies on a greater number of sophisticated, reliable, and quiet motors. Furthermore, the inherent quietness of a BEV's electric powertrain makes other sources of noise, such as the operation of seat motors, far more perceptible [2, 5]. This has placed a significant emphasis on Noise, Vibration, and Harshness (NVH) performance, compelling manufacturers to develop motors that operate almost silently. Finally, in a market where vehicle range is a paramount concern, the energy efficiency of every component, no matter how small, contributes to the overall system optimization [17, 62]. While the power consumption of seat motors is intermittent and relatively low compared to the main traction motor, the cumulative effect of numerous auxiliary systems is a key consideration for automotive engineers. This report will provide a comprehensive examination of the technical specifications, performance metrics, and supplier landscape for the electric seat motors defining the passenger experience in 2024-2026 European and Asian BEV models.

2.0 Motor Types and Functional Integration

The actuation of power seats in modern BEVs is accomplished through a system of small DC motors, control units, and mechanical linkages. The technology is primarily centered around two main types of DC motors, with a third type used for highly specific applications. The most common are **brushed DC motors**, which have long been the industry standard due to their simple construction, low cost, and the ease with which their speed and torque can be controlled [5, 62]. However, the mechanical brushes used to transfer power to the rotor are subject to wear over time and can generate electrical noise and audible sound [62]. In response to the demand for greater longevity and quieter operation, especially in premium vehicles, the industry is increasingly adopting **Brushless DC (BLDC) motors**. BLDC motors use electronic commutation instead of mechanical brushes, resulting in significantly higher efficiency, a longer operational lifespan, reduced maintenance requirements, and much quieter performance [5, 16, 62]. Their superior characteristics make them the preferred choice for luxury BEVs where NVH performance and reliability are critical [5, 16]. A third, less common type, the **stepper motor**, is sometimes employed for applications requiring extremely precise positional control without the need for a feedback sensor, though their use in primary seat adjustments is less prevalent than brushed or brushless DC motors [5].

The number of motors integrated into a single car seat directly correlates with its level of adjustability and the vehicle's market segment. In standard or entry-level BEV models, a seat may be equipped with two to four motors to provide basic 4-way or 8-way adjustments [11, 19]. A typical 6-way adjustable seat, common in mass-market cars, generally uses three motors: one for fore-aft movement, one for height adjustment, and one for the backrest recline [11]. An 8-way seat adds a fourth motor for powered lumbar support [11]. In contrast, premium European and Asian BEVs offer seats with a much higher degree of articulation, often marketed as 12-way, 16-way, or even 20-way adjustable. These advanced systems can employ six or more motors per seat [16, 19]. Additional motors are dedicated to functions such as adjusting the length of the seat cushion, controlling the lateral support of the side bolsters, positioning the headrest, and actuating complex multi-zone lumbar and massage systems [11, 16]. These premium systems almost exclusively utilize BLDC motors to ensure smooth, quiet, and reliable operation, managed by a dedicated seat control unit that coordinates the movements and often includes memory functions to store preferred positions for multiple drivers [16, 19].

3.0 Core Electrical and Performance Specifications

The operational parameters of electric seat motors are defined by a set of core electrical and performance specifications that dictate their power, strength, and control. These motors are designed to operate on the vehicle's low-voltage DC electrical system, which is typically a nominal **12-volt** system [7, 12]. While some larger commercial vehicles may use 24V systems, and future vehicle architectures

could explore higher auxiliary voltages, the 12V standard remains dominant for passenger BEVs in the 2024-2026 timeframe. The power consumption of these motors is a function of the current they draw. The **current draw (amperage)** varies significantly depending on the motor's operational state. A motor under no load may draw a relatively low current, often less than 3 amperes. For instance, a specific Bosch seat motor model has a specified no-load current of 3A or less [32, 41]. However, when the motor is under load, moving the seat and its occupant, the current draw increases. A typical motor might draw between 4 and 7 amps during normal use [9]. The maximum current is drawn when the motor is stalled—for example, when the seat reaches the end of its travel track or is obstructed. This **stall current** can be substantial, with some motors rated for 12A, and others potentially drawing up to 20A [10, 32, 41]. Given that a premium seat can have multiple motors operating simultaneously, the total potential current draw can be significant, necessitating robust wiring and circuit protection, typically in the form of a 25-amp or 30-amp fuse or circuit breaker for the power seat system [7, 10].

In terms of power, the **wattage** of a single seat motor is generally modest. A specific model from Nidec is rated at an output of 60 watts [15], while a popular Bosch motor is rated at 44 watts [32, 41]. These figures represent typical operating power, as the theoretical power at stall (Voltage x Stall Current) can be much higher, for example, 144 watts for a 12V motor stalling at 12A [32, 41]. The mechanical performance is defined by torque and speed. The high-speed, low-torque rotation of the motor's shaft is converted into low-speed, high-torque motion by an integrated gearbox. These gearboxes, often using worm-drive or planetary gear arrangements, are critical for providing the force needed to move the seat mechanism against the weight of an occupant and for preventing the seat from moving when the motor is not powered (a property known as resistance to backdrive) [14, 32]. A specific Bosch motor, for example, features an internal gear ratio of 174.9:1 [41], allowing it to produce a formidable **stall torque** of 22 Newton-meters (16.2 ft-lb) [32, 41]. The resulting output **speed** is slow and controlled, with the same Bosch motor having a no-load free speed of just 24 revolutions per minute (RPM) [32, 41]. For precise control and the implementation of memory functions, many modern seat motors incorporate **Hall effect sensors**. These sensors provide rotational feedback to the seat's electronic control unit (ECU), allowing it to track the exact position of the seat for accurate, repeatable adjustments [32].

4.0 Physical Design, Weight, and Acoustic Characteristics

The physical design of automotive seat motors is governed by the dual constraints of limited space within the seat structure and the overarching industry-wide push for vehicle lightweighting. To improve energy efficiency and extend the range of BEVs, manufacturers seek to reduce the mass of every component. Consequently, suppliers are focused on the **miniaturization** of seat motors, developing more compact and power-dense designs [14, 23]. This is often achieved by integrating the motor and its associated gearbox into a single, space-saving unit. The weight of an individual motor is relatively low; for example, a specific automotive seat adjuster motor from Nidec has a mass of just 270 grams (0.6 lbs) [15], and a geared motor from Bosch weighs 363 grams (0.8 lbs) [41]. While these individual weights are small, a premium seat with numerous motors and heavy-duty adjustment mechanisms can contribute significantly to the vehicle's overall mass, with a complete power seat assembly weighing 70 pounds or more [13]. Suppliers like Brose have achieved weight savings by using high-strength steels and innovative material combinations in their seat structures and adjuster components [55].

The materials used in the construction of the motors and their gearing are chosen to balance durability, weight, and cost. While motor housings are typically metal, internal gears may be made of high-strength plastics to reduce weight and noise. However, this can be a point of failure if the mechanism is subjected to sudden external shocks [53]. The design of the output interface is also a key considera-

tion, with manufacturers often using splined or star-shaped outputs to ensure a robust connection to the seat's mechanical linkages [32].

A critical aspect of motor design for BEVs is **acoustics**, or NVH (Noise, Vibration, and Harshness) performance. The absence of a loud internal combustion engine in a BEV makes other, previously masked sounds much more prominent [2, 5]. The whirring or grinding of a seat motor can be perceived as a sign of low quality and can detract from the premium experience automakers aim to create. Major suppliers invest heavily in optimizing the acoustic performance of their motors. Brose, for instance, highlights its brushless DC motors with integrated electronics for their excellent acoustics and variable adjustment speeds, which can be controlled to produce a more harmonious and uniform sound [16, 29]. The shift towards brushless DC motors is a key strategy in this area, as their non-contact electronic commutation inherently produces less noise and vibration than the friction-based system of their brushed counterparts [5, 62].

5.0 Leading Suppliers and Product Specifications

The market for automotive seat motors is dominated by a handful of global mechatronics specialists who supply components directly to original equipment manufacturers (OEMs). The specifications of their products provide a clear insight into the technology being deployed in 2024-2026 European and Asian BEVs.

Robert Bosch GmbH (Germany): A leading European supplier, Bosch provides a wide range of seat drives. A well-documented example is the model 6 004 RA3 194-06, often used in various automotive applications. This 12V DC motor is characterized by a no-load current of 3A or less and a maximum stall current of 12A, with a rated power consumption of 44W [32, 41]. It features a high-ratio worm-driven gear system (174.9:1) that delivers a substantial stall torque of 22 Nm at a low output speed of 24 RPM [32, 41]. Weighing 363g, it incorporates an integrated Hall sensor for precise positioning feedback and a thermal switch for overload protection [32, 41]. Bosch emphasizes compact, lightweight designs that combine the motor and gear unit into a single item [14].

Brose Fahrzeugteile SE & Co. KG (Germany): Brose is a key competitor, specializing in complete mechatronic seat systems [54]. The company is a major proponent of highly efficient brushless DC motors with integrated electronics, which are central to their next-generation seat adjusters [16, 29]. These motors offer superior acoustics, a longer lifetime, and higher power density [16]. Brose's modular seat structures can accommodate up to 20-way power adjustments and are designed with a focus on lightweight construction [16, 55]. Their seat control units can manage up to 10 motors, enabling features like anti-trap protection and speed-controlled operation to optimize acoustics, making them a preferred supplier for premium European automakers [16].

Nidec Corporation (Japan): As a global motor manufacturing giant, Nidec offers a range of automotive seat adjuster motors [50, 56]. A specific mid-sized DC motor ($\phi 41$ model) from their portfolio operates at 13V, drawing 12.5A to produce a 60W output [15]. It has a mass of 270g and delivers a torque of 300 mN·m, with a stall torque of 650 mN·m (note: this torque is likely pre-gearing) [15]. Nidec's motors are engineered for long life, high power, quiet operation, and low power consumption, aligning with the core requirements of the BEV market [15]. The company is actively developing even more compact motor ranges to further reduce weight and improve packaging density [50].

Johnson Electric (Hong Kong/China): A major Asian supplier with a strong global presence, Johnson Electric's Automotive Products Group (APG) provides a wide array of motion solutions for vehicle interiors, including high-power seat adjustment motors [36, 63]. Their products are noted for their compact design, low weight, and excellent NVH and Electromagnetic Compatibility (EMC) performance [36]. Having acquired the seat motor business of ArvinMeritor, Johnson Electric has deep expertise in this

segment, offering customized solutions for seat adjustment, lumbar support, and headrest control to major automakers worldwide [37].

Mabuchi Motor (Japan): Mabuchi is another dominant Japanese player in the small electric motor market. The company specifically recommends models like the RZ-46BWA and SZ-286WA for power seat applications [51]. Mabuchi's market strategy is aligned with key industry trends, including the development of smart, lightweight, and energy-efficient motors [51, 66]. Their products are integral to the supply chain for many Asian vehicle manufacturers, and they compete on cost and volume, particularly in the passenger car segment.

These suppliers, along with others like Lear Corporation, Forvia, and Adient who focus more on complete seat systems, form the technological backbone of modern automotive seating, providing the essential motion components for BEVs across all market segments [22, 23, 24].

6.0 Conclusion: The Future of Seat Actuation in BEVs

The specifications of electric seat motors for 2024-2026 BEV models reflect a clear and decisive technological evolution shaped by the unique characteristics of electric mobility and rising consumer expectations for comfort and sophistication. The analysis of products from leading suppliers and prevailing market trends reveals a distinct bifurcation in technology based on vehicle segment. Standard-tier BEVs will continue to utilize proven, cost-effective brushed DC motors to provide essential 6-way or 8-way adjustments, prioritizing functionality and value [5, 19, 62]. In stark contrast, the premium BEV segment is defined by the widespread adoption of advanced Brushless DC (BLDC) motors. These quieter, more efficient, and more durable motors are essential for powering the complex, multi-motor systems that enable 12-way or greater adjustments, dynamic massage functions, and seamless integration with memory and personalization features [5, 16, 18].

The technical landscape is characterized by 12V DC motors engineered for high torque output via integrated gearboxes, with typical power ratings of 40-60W and stall currents exceeding 12A [15, 32, 41]. The push for vehicle efficiency has driven a focus on lightweighting and miniaturization, with individual motors weighing as little as 270 grams [15]. Perhaps most critically for the BEV environment, acoustic performance has become a primary design driver, compelling suppliers to deliver motors that operate with near-imperceptible noise levels [5, 16].

Looking forward, the trajectory is toward even greater integration and intelligence. Seat motors will not be standalone components but integral parts of a smart, mechatronic system that combines mechanical actuation with advanced sensors and software [5, 16, 65]. This will enable new levels of safety, comfort, and personalization, further transforming the vehicle interior into an adaptive, user-centric space. The quiet, efficient, and intelligent electric seat motor is, therefore, not merely a component of convenience but a key enabler of the next generation of in-vehicle experience.

References

1. [Alternative Fuels Data Center: How Do All-Electric Cars Work? - Alternative Fuels Data Center](https://afdc.energy.gov/vehicles/how-do-all-electric-cars-work) (<https://afdc.energy.gov/vehicles/how-do-all-electric-cars-work>)
2. [Electric Motor \(BEV\) – Function & Power Delivery | Mein Autolexikon - My-cardictionary.com](https://www.my-cardictionary.com/dictionary/drive-system-bev/electric-motor-bev) (<https://www.my-cardictionary.com/dictionary/drive-system-bev/electric-motor-bev>)
3. [Electric and Hybrid Vehicles: Battery, Charging & Safety | NHTSA - Nhtsa.gov](https://www.nhtsa.gov/vehicle-safety/electric-and-hybrid-vehicles) (<https://www.nhtsa.gov/vehicle-safety/electric-and-hybrid-vehicles>)

4. [What's the difference between BEV and EV? - ShanDong TaiRui - TaiRui Auto](https://www.tairuiauto.com/news/whats-the-difference-between-bev-and-ev/) (https://www.tairuiauto.com/news/whats-the-difference-between-bev-and-ev/)
5. [Battery Electric Vehicle \(BEV\)-Everything you need to know - MechBasic](https://mechbasic.com/battery-electric-vehicle-bev-everything-you-need-to-know/) (https://mechbasic.com/battery-electric-vehicle-bev-everything-you-need-to-know/)
6. [BEV, PHEV or HEV: The Differences Affect the Architecture - Aptiv](https://www.apativ.com/en/insights/article/bev-phev-or-hev-the-differences-affect-the-architecture) (https://www.apativ.com/en/insights/article/bev-phev-or-hev-the-differences-affect-the-architecture)
7. [power carseat from 240v. - diynot.com](https://www.diynot.com/diy/threads/power-carseat-from-240v.418434/) (https://www.diynot.com/diy/threads/power-carseat-from-240v.418434/)
8. [Heated Seat Power Consumption Measurements - gm-volt.com](https://www.gm-volt.com/threads/heated-seat-power-consumption-measurements.10064/) (https://www.gm-volt.com/threads/heated-seat-power-consumption-measurements.10064/)
9. [Power seats question. - bobistheoilguy.com](https://bobistheoilguy.com/forums/threads/power-seats-question.358529/) (https://bobistheoilguy.com/forums/threads/power-seats-question.358529/)
10. [Amp rating for electric seat adjustment? - cherokeeforum.com](https://www.cherokeeforum.com/f2/amp-rating-electric-seat-adjustment-248192/) (https://www.cherokeeforum.com/f2/amp-rating-electric-seat-adjustment-248192/)
11. [Power Adjustable Driver Seat Feature Explained - v3cars.com](https://www.v3cars.com/features-explained/power-adjustable-driver-seat-feature-explained) (https://www.v3cars.com/features-explained/power-adjustable-driver-seat-feature-explained)
12. [How to power an electric BMW seat - e46fanatics.com](https://www.e46fanatics.com/threads/how-to-power-an-electric-bmw-seat.806966/) (https://www.e46fanatics.com/threads/how-to-power-an-electric-bmw-seat.806966/)
13. [how much do seat motors weigh? - vwvortex.com](https://www.vwvortex.com/threads/how-much-do-seat-motors-weigh.1775782/) (https://www.vwvortex.com/threads/how-much-do-seat-motors-weigh.1775782/)
14. [Seat drive - bosch-mobility.com](https://www.bosch-mobility.com/en/solutions/actuators/seat-drive/) (https://www.bosch-mobility.com/en/solutions/actuators/seat-drive/)
15. [Automotive seat adjuster - nidec.com](https://www.nidec.com/en/product/search/category/B101/M102/S100/NCJ-CM4/) (https://www.nidec.com/en/product/search/category/B101/M102/S100/NCJ-CM4/)
16. [Adjustment systems for front and rear seats - brose.com](https://www.brose.com/de-en/products/adjustment-systems-for-front-and-rear-seats/) (https://www.brose.com/de-en/products/adjustment-systems-for-front-and-rear-seats/)
17. [Automotive Seat Adjustment Motor Market - globalgrowthinsights.com](https://www.globalgrowthinsights.com/market-reports/automotive-seat-adjustment-motor-market-106168) (https://www.globalgrowthinsights.com/market-reports/automotive-seat-adjustment-motor-market-106168)
18. [Motors for Seat Actuation - portescap.com](https://www.portescap.com/en/industries-supported/aerospace/motors-for-seat-actuation) (https://www.portescap.com/en/industries-supported/aerospace/motors-for-seat-actuation)
19. [Power seat - grokipedia.com](https://gropedia.com/page/Power_seat) (https://gropedia.com/page/Power_seat)
20. [one-motor power seat - toyota-boshoku.com](https://www.toyota-boshoku.com/global/development/product_technology/power-seat/) (https://www.toyota-boshoku.com/global/development/product_technology/power-seat/)
21. [Are Powered Seats Useful? - gomechanic.in](https://gomechanic.in/blog/are-powered-seats-useful/) (https://gomechanic.in/blog/are-powered-seats-useful/)
22. [Automotive Electric Seats Market Size & Share, Trends, Growth Forecast - arizton.com](https://www.arizton.com/market-reports/automotive-electric-seats-market) (https://www.arizton.com/market-reports/automotive-electric-seats-market)
23. [Global and China Automotive Seat Motor Markets, 2020-2026 - markets.businessinsider.com](https://markets.businessinsider.com/news/stocks/global-and-china-automotive-seat-motor-markets-2020-2026-1030118884) (https://markets.businessinsider.com/news/stocks/global-and-china-automotive-seat-motor-markets-2020-2026-1030118884)
24. [In-Depth Industry Outlook: Automotive Seat Motor Market Size, Forecast - verifiedmarketresearch.com](https://www.verifiedmarketresearch.com/product/automotive-seat-motor-market/) (https://www.verifiedmarketresearch.com/product/automotive-seat-motor-market/)
25. [Automotive Power Seat Motor Market Research Report 2032 - dataintelo.com](https://dataintelo.com/report/global-automotive-power-seat-motor-market) (https://dataintelo.com/report/global-automotive-power-seat-motor-market)
26. [Automotive Seat Motor Market Size, Share & Forecast to 2030 - researchandmarkets.com](https://www.researchandmarkets.com/report/automotive-seat-motor) (https://www.researchandmarkets.com/report/automotive-seat-motor)
27. [Automotive Seat Motor Market Outlook 2026-2034 - intelmarketresearch.com](https://www.intelmarketresearch.com/global-automotive-seat-motor-forecast-market-26564) (https://www.intelmarketresearch.com/global-automotive-seat-motor-forecast-market-26564)
28. [Brose - Products and innovative solutions - brose.com](https://www.brose.com/de-en/products/) (https://www.brose.com/de-en/products/)

29. [Electric motors and drives - brose.com](https://www.brose.com/de-en/products/electric-motors-and-drives/) (https://www.brose.com/de-en/products/electric-motors-and-drives/)
30. [Brose in the Czech Republic - Press - brose.com](https://www.brose.com/cz-en/brose-in-cz/press/2021/pressdetailpages2021_366275.html) (https://www.brose.com/cz-en/brose-in-cz/press/2021/pressdetailpages2021_366275.html)
31. [Brose Fahrzeugteile - marklines.com](https://www.marklines.com/en/top500/brose-fahrzeugteile) (https://www.marklines.com/en/top500/brose-fahrzeugteile)
32. [Bosch Seat Motor - wpilib.screenstepslive.com](https://wpilib.screenstepslive.com/s/currentCS/m/kop/l/679357-bosch-seat-motor) (https://wpilib.screenstepslive.com/s/currentCS/m/kop/l/679357-bosch-seat-motor)
33. [2017 Bosch Seat Motor - chiefdelphi.com](https://www.chiefdelphi.com/t/2017-bosch-seat-motor/154692) (https://www.chiefdelphi.com/t/2017-bosch-seat-motor/154692)
34. [Bosch Seat Motor and Harness Cable - andymark.com](https://www.andymark.com/products/bosch-seat-motor-and-harness-cable) (https://www.andymark.com/products/bosch-seat-motor-and-harness-cable)
35. [Bosch Seat Motor - dekuprobotics.com](https://dekuprobotics.com/en_US/product/bosch-seat-motor?type=1462) (https://dekuprobotics.com/en_US/product/bosch-seat-motor?type=1462)
36. [Automotive Products Group - pdf.directindustry.com](https://pdf.directindustry.com/pdf/johnson-electric/automotive-products-group/665-1042405.html) (https://pdf.directindustry.com/pdf/johnson-electric/automotive-products-group/665-1042405.html)
37. [Johnson Electric Machinery Factory Co., Ltd. \(Formerly Johnson Electric Group\) - MarkLines Automotive Industry Portal - marklines.com](https://www.marklines.com/en/top500/johnson-electric-machinery-factory) (https://www.marklines.com/en/top500/johnson-electric-machinery-factory)
38. [Johnson Electric - Wikipedia - en.wikipedia.org](https://en.wikipedia.org/wiki/Johnson_Electric) (https://en.wikipedia.org/wiki/Johnson_Electric)
39. [Motors - shop.johnsonelectric.com](https://shop.johnsonelectric.com/collections/motors/motor) (https://shop.johnsonelectric.com/collections/motors/motor)
40. [What's the difference between BEV and EV? - tairuiauto.com](https://www.tairuiauto.com/news/whats-the-difference-between-bev-and-ev/) (https://www.tairuiauto.com/news/whats-the-difference-between-bev-and-ev/)
41. [FRC Bosch motor 6 004 RA3 194-06 spec sheetv1.pdf - files.andymark.com](http://files.andymark.com/FRC_Bosch_motor_6_004_RA3_194-06_spec_sheetv1.pdf) (http://files.andymark.com/FRC_Bosch_motor_6_004_RA3_194-06_spec_sheetv1.pdf)
42. [Bosch Seat Motor and Harness Cable - dekuprobotics.com](https://dekuprobotics.com/en_US/product/bosch-seat-motor-and-harness-cable?type=1529) (https://dekuprobotics.com/en_US/product/bosch-seat-motor-and-harness-cable?type=1529)
43. [2016-12-21 spec sheet - Bosch FRC motor 6 004 RA3 194-06.pdf - cdn.andymark.com](https://cdn.andymark.com/media/W1siZiIsIjIwMTkvMDEvMjIvMTUvNDAvMDgvNDE1ZWE5ZmUtZjA4N-i00M2MxLWE1YjMtYTlhNzJhZWE0NmMxLzlwMTYtMTItMjEgc3BIYyBzaGVldCAtIEJvc2NoIEZSQyBtb3RvcjA2IDAuNCBSQTMgMTk0LTA2LnBkZjIjXQ/2016-12-21%20spec%20sheet%20-%20Bosch%20FRC%20motor%206%20004%20RA3%20194-06.pdf?sha=42a181c2a13b270e) (https://cdn.andymark.com/media/W1siZiIsIjIwMTkvMDEvMjIvMTUvNDAvMDgvNDE1ZWE5ZmUtZjA4N-i00M2MxLWE1YjMtYTlhNzJhZWE0NmMxLzlwMTYtMTItMjEgc3BIYyBzaGVldCAtIEJvc2NoIEZSQyBtb3RvcjA2IDAuNCBSQTMgMTk0LTA2LnBkZjIjXQ/2016-12-21%20spec%20sheet%20-%20Bosch%20FRC%20motor%206%20004%20RA3%20194-06.pdf?sha=42a181c2a13b270e)
44. [ERMEC, S.L. BARCELONA Tel.: \(+34\) 902 450 160 ERMEC, S.L. MADRID PORTUGAL - ermec.com](http://www.ermec.com/johnsonelectric/catalogo/2007/JE_Motor.pdf) (http://www.ermec.com/johnsonelectric/catalogo/2007/JE_Motor.pdf)
45. [Johnson DC Motors - Johnson Electric - PDF Catalogs | Technical Documentation | Brochure - pdf.directindustry.com](https://pdf.directindustry.com/pdf/johnson-electric/johnson-dc-motors/665-6344.html) (https://pdf.directindustry.com/pdf/johnson-electric/johnson-dc-motors/665-6344.html)
46. [SAIA Motor Catalog Johnson Electric Motors - johnsonelectric.com](https://www.johnsonelectric.com/pub/media/wysiwyg/je/je_pdf/saia_motor_catalog.pdf) (https://www.johnsonelectric.com/pub/media/wysiwyg/je/je_pdf/saia_motor_catalog.pdf)
47. [Adjustment systems for front and rear seats - brose.com](https://www.brose.com/us-en/products/adjustment-systems-for-front-and-rear-seats/) (https://www.brose.com/us-en/products/adjustment-systems-for-front-and-rear-seats/)
48. [Brose - Excellence in Mechatronics - brose.com](https://www.brose.com/de-en/) (https://www.brose.com/de-en/)
49. [Nidec Develops New Automotive Air Suspension Motor - nidec.com](https://www.nidec.com/en/product/news/2024/news0603-02/) (https://www.nidec.com/en/product/news/2024/news0603-02/)
50. [Product Line Automotive Seat Adjusters - nidec-ma.de](http://www.nidec-ma.de/en/automotiveseat-adjusters.html) (http://www.nidec-ma.de/en/automotiveseat-adjusters.html)
51. [Automotive - mabuchi-motor.com](https://www.mabuchi-motor.com/product/case/automotive.html) (https://www.mabuchi-motor.com/product/case/automotive.html)

52. [Mabuchi RZ-46BWA 12-24V Automobile Seat Adjustment Motor - aliexpress.com](https://www.aliexpress.com/item/32239696860.html) (https://www.aliexpress.com/item/32239696860.html)
53. [Bosch Seat motor 6004RA3194-06 - Motors - Chief Delphi - chiefdelphi.com](https://www.chiefdelphi.com/t/bosch-seat-motor-6004ra3194-06/162320) (https://www.chiefdelphi.com/t/bosch-seat-motor-6004ra3194-06/162320)
54. [Brose Sitech - brose-sitech.com](https://www.brose-sitech.com/en) (https://www.brose-sitech.com/en)
55. [Brose produces five millionth lightweight seat structure for Daimler and BMW - brose.com](https://www.brose.com/de-en/press/2016/brose-produces-five-millionth-lightweight-seat-structure-for-daimler-and-bmw.html) (https://www.brose.com/de-en/press/2016/brose-produces-five-millionth-lightweight-seat-structure-for-daimler-and-bmw.html)
56. [Automotive | NIDEC CORPORATION - nidec.com](https://www.nidec.com/en/product/search/category/B105/) (https://www.nidec.com/en/product/search/category/B105/)
57. [What is Competitive Landscape of Johnson Electric Holdings Company? – PortersFiveForce.com - portersfiveforce.com](https://portersfiveforce.com/blogs/competitors/johnsonelectric) (https://portersfiveforce.com/blogs/competitors/johnsonelectric)
58. [Design and Experimental Study of an Active Seat Suspension Based on a Rotating Motor and Gear Reducer - pmc.ncbi.nlm.nih.gov](https://pmc.ncbi.nlm.nih.gov/articles/PMC9371391/) (https://pmc.ncbi.nlm.nih.gov/articles/PMC9371391/)
59. [Automotive Power Seat Actuator Motor Market - wiseguyreports.com](https://www.wiseguyreports.com/reports/automotive-power-seat-actuator-motor-market) (https://www.wiseguyreports.com/reports/automotive-power-seat-actuator-motor-market)
60. [Automotive Seat Motor Market Size and Forecast - verifiedmarketreports.com](https://www.verifiedmarketreports.com/product/automotive-seat-motor-market-size-and-forecast/) (https://www.verifiedmarketreports.com/product/automotive-seat-motor-market-size-and-forecast/)
61. [Micro Motors - Miniature and small DC Micro Motors - firgelliauto.com](https://www.firgelliauto.com/collections/micro-motors) (https://www.firgelliauto.com/collections/micro-motors)
62. [Automotive Small DC Motors Market Size & Share Report, 2031 - metastatinsight.com](https://www.metastatinsight.com/report/automotive-small-dc-motors-market) (https://www.metastatinsight.com/report/automotive-small-dc-motors-market)
63. [Automotive Compact DC Motors - johnsonelectric.com](https://www.johnsonelectric.com/en/products/dc-motors/dcmotor1) (https://www.johnsonelectric.com/en/products/dc-motors/dcmotor1)
64. [190 Rpm 12Vdc Right Angle Drive Electric Motor From Car Seats - amazon.com](https://www.amazon.com/12Vdc-Right-Angle-Drive-Electric/dp/B005IR1NBA) (https://www.amazon.com/12Vdc-Right-Angle-Drive-Electric/dp/B005IR1NBA)
65. [Brose offers mechatronics expertise - Brose](https://www.brose.com/de-en/products/seating-systems/) (https://www.brose.com/de-en/products/seating-systems/)
66. [Products Information - Mabuchi Motor](https://www.mabuchi-motor.com/product/) (https://www.mabuchi-motor.com/product/)